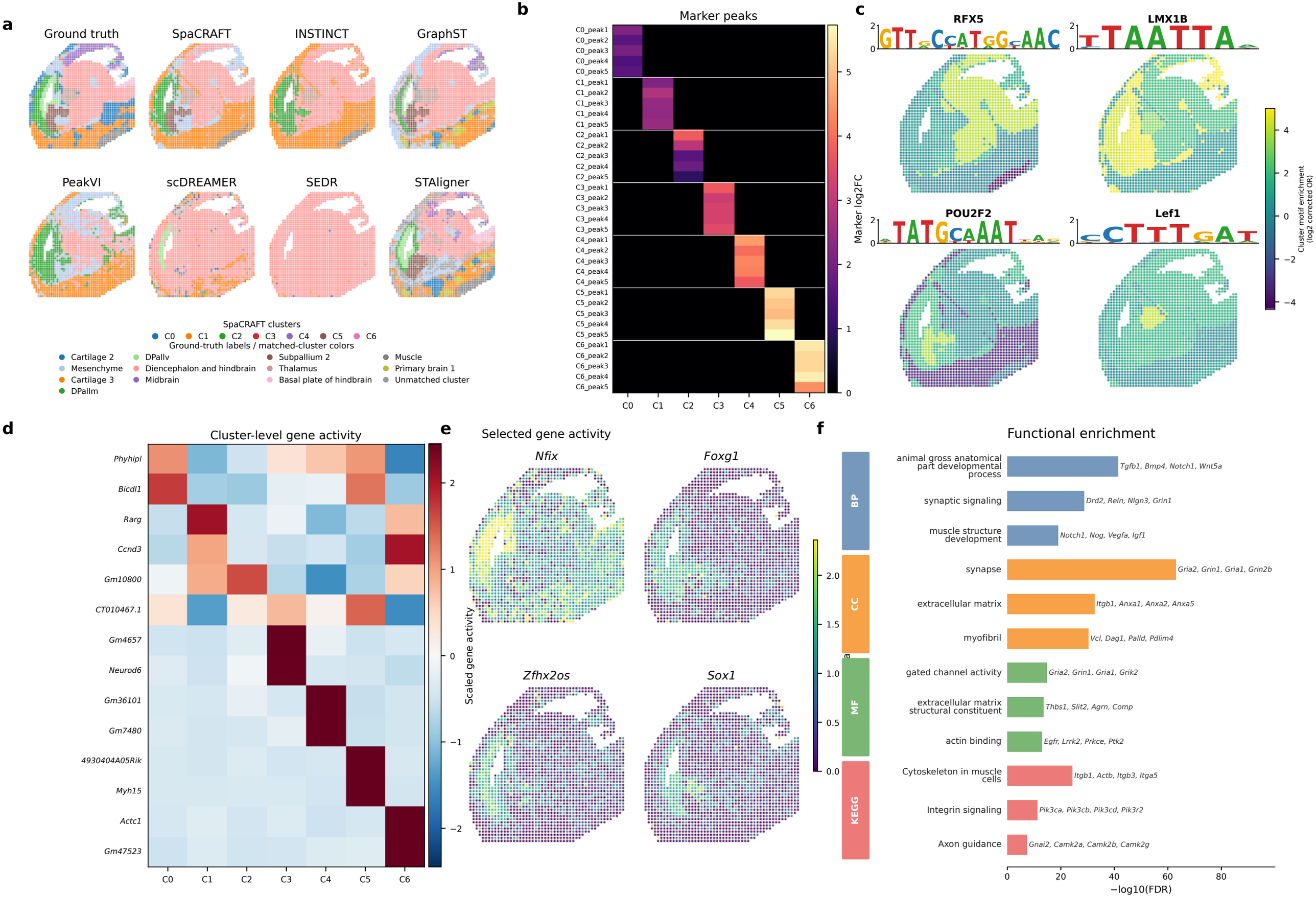

a
Ground truth
SpaCRAFT
INSTINCT
GraphST
PeakVI
scDREAMER
SEDR
STAligner
SpaCRAFT clusters
C0 C1 C2 C3 C4 C5 C6
Ground-truth labels / matched-cluster colors
Cartilage 2
Mesenchyme
Cartilage 3
DPallm
DPallv
Diencephalon and hindbrain
Midbrain
Subpallium 2
Thalamus
Basal plate of hindbrain
Muscle
Primary brain 1
Unmatched cluster
b
Marker peaks
Marker log2FC
c
RFX5
LMX1B
POU2F2
Lef1
Cluster motif enrichment (log2 corrected OR)
d
Cluster-level gene activity
Phyhipl
Bicdl1
Rarg
Ccnd3
Gm10800
CT010467.1
Gm4657
Neurod6
Gm36101
Gm7480
4930404A05Rik
Myh15
Actc1
Gm47523
e
Selected gene activity
Nfix
Foxg1
Zfhx2os
Sox1
Scaled gene activity
f
Functional enrichment
BP
CC
MF
KEGG
animal gross anatomical part developmental process
Tgfb1, Bmp4, Notch1, Wnt5a
synaptic signaling
Drd2, Reln, Nlgn3, Grin1
muscle structure development
Notch1, Nog, Vegfa, Igf1
synapse
Gria2, Grin1, Gria1, Grin2b
extracellular matrix
Itgb1, Anxa1, Anxa2, Anxa5
myofibril
Vcl, Dag1, Palld, Pdlim4
gated channel activity
Gria2, Grin1, Gria1, Grik2
extracellular matrix structural constituent
Thbs1, Slit2, Agrn, Comp
actin binding
Egfr, Lrrk2, Prkce, Ptk2
Cytoskeleton in muscle cells
Itgb1, Actb, Itgb3, Itga5
Integrin signaling
Pik3ca, Pik3cb, Pik3cd, Pik3r2
Axon guidance
Gnai2, Camk2a, Camk2b, Camk2g
−log10(FDR)